

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 3 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: BTCVC303 - BUILDING CONSTRUCTION & DRAWING

Time : 4 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 The lengthy face of the brick is known as
A. Face B. King closer C. Queen closer D. Stretcher
- 2 The highest central wedge-shaped block of an arch is called..

A. Haunch B. Key Stone C. Skewback D. Spandril
- 3 The process of taking out stone from natural rock is known as
A. Weathering B. Quarrying C. Dressing D. Cladding
- 4 A flat arch is suitable for span up to ..
A. 1.5 M B. 0.5 M C. 1.00 M D. 1.25 M
- 5 The volume of one bag of cement is
A. 0.0214 cu. m B. 0.0347 cu. m C. 0.0434 cu. m D. .0609 cu. m
- 6 The process of rolling, folding, and spreading of the particle is known as the _____ of concrete.

A) Curing B) Mixing C) Grouting D) Watering
- 7 The vertical portion between each tread on the stairs is called...
A. Going B. Nosing C. Winder D. Riser
- 8 A baluster in a stair is the _____ member supporting _____.
A. Horizontal, handrail B. Vertical, landing C. Vertical, handrail
D. Horizontal, landing
- 9 A sloping roof formed by four sloping surfaces in four directions is known as....
A. Gabled Roof B. Mansard Roof C. Hip Roof D. Grambled Roof
- 10 The couple roof is used for a span of...
A. 3.5 m or less B. 3.5 m but less than 5.0 m
C. 5 m to 6.5 m D. 6.5 m to 8.0 m
- 11 What is the primary advantage of using the rat trap bond construction technique in building construction?
A. It reduces the amount of mortar needed for construction
B. It increases the structural stability of the walls
C. It speeds up the construction process
D. It enhances thermal insulation properties

- 12 What is a key feature of energy-efficient roofing systems?
 A. They are primarily made of metal
 B. They utilize reflective materials to reduce heat absorption
 C. They require frequent maintenance to remain efficient
 D. They are more expensive than traditional roofing materials
- Q2. Solve the following.
- A) Describe general principles to be observed in stone masonry construction. 6
- B) Sketch a semicircular arch, illustrating all the technical terms used in arch. Define six five components. 6
- Q3. Solve the following.
- A) Define the workability and explain the slump cone test with a neat sketch. <https://www.batuonline.com> 6
- B) Define formwork. Explain the different types of formwork. 6
- Q4. Solve Any One of the following.
- A) Draw a neat sketch of a dog-legged staircase showing all components. 12
- B) Draw a neat sketch of a Framed and Panelled door? (Plan , Section & Elevation) 12
- Q5. Solve Any One of the following.
- A) Explain different types of flooring. 12
- B) Draw a neat sketch of King post truss and its joint. 12
- Q6. Solve Any One of the following.
- A) Describe various guidelines for transportation and erection in prefabrication. 12
- B) Explain the term 'Tolerance 'in prefabrication. 12

*** End ***